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ORIGINAL ARTICLE

Deciphering terrorist attributions: a global terrorism data-driven comparison of machine learning models

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Abstract

Attributing terrorist incidents to responsible groups remains a critical challenge in counter-terrorism, complicated by the evolving complexity and decentralization of extremist operations. This study addresses the urgent need for automated, accurate, and scalable attribution mechanisms by proposing and rigorously evaluating two novel ensemble-based machine learning models: an adaptive weighted voting ensemble (WVE) and a hierarchical Stacking Classifier with optimized meta-learner selection. Leveraging structured incident data from the Global Terrorism Database, the research implements a comprehensive methodological framework encompassing domain-specific feature engineering, advanced class imbalance handling through SMOTE-Tomek, and systematic hyperparameter optimization. Robust nested cross-validation is employed throughout to ensure unbiased model selection and reliable performance estimation. The study conducts comparative analysis across ten established machine learning algorithms and the two proposed ensemble methods. Experimental results demonstrate that the Stacking Classifier achieves 73.70% accuracy ($\pm$ 1.23) and an AUC-ROC of 0.892, while the WVE attains 73.38% accuracy ($\pm$ 1.18) and an AUC-ROC of 0.889-representing a 15–20% improvement over conventional methods. Both models maintain robust precision-recall balance, with their superiority statistically validated ($p < 0.01$, Friedman-Nemenyi test) in multi-class attribution tasks; furthermore, integrated SHAP analysis (showing 94% correlation with traditional importance) enables transparent, instance-level interpretation essential for operational deployment in counter-terrorism contexts. The proposed framework advances the field by offering interpretability, operational scalability, and multi-dimensional attribution capabilities, equipping security agencies with effective tools for real-time threat identification and response.

Keywords Machine learning ensembles · Terrorism attribution · Feature engineering · Counter-terrorism analytics · Multi-class classification · Global terrorism database

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1 Introduction

The global terrorism environment has become a multi-dimensional development involving political, ideological, and religious inspirations, posing unprecedented national security and global peace challenges (Renard and de Kerchove, 2021). Contemporary terrorist groups, especially decentralized networks such as ISIS and Al-Qaeda, function through loosely coupled self-contained subgroups, complicating attack attribution (Black et al. 2022; Haghani et al. 2022; Khan et al. 2023). The global terrorism database (GTD) lists more than 109,000 terrorist attacks from the period 2000–2020 that resulted in over 300,000 fatalities, highlighting the urgent necessity of threat identification and prevention based on accurate assessments (Jaspersen and Montibeller 2020).

Traditional terrorist group attribution methods face significant limitations due to covert organizational structures, proxy actor utilization, and dynamic operational strategies (Adigwe et al. 2024). Digital platforms have further complicated attribution processes, as terrorists exploit anonymous channels and disseminate disinformation (Feng et al. 2020; Bolshibayeva et al. 2023). Conventional human expert-based classification is time-intensive, bias-prone, and struggles to keep pace with evolving terrorist behaviors (Alazab et al. 2023). Machine learning approaches offer promising solutions for enhancing attribution accuracy through pattern identification in large-scale terrorism datasets. However, existing applications face challenges including limited comparative algorithm analysis, insufficient handling of class imbalances, lack of interpretability in security-critical applications, and absence of robust ensemble methods specifically designed for terrorism attribution tasks (Montasari 2024).

Given these limitations in existing approaches, this research addresses the fundamental question: How can ensemble machine learning methods be optimized to enhance multi-class terrorist group attribution accuracy and reliability using structured incident data from the Global Terrorism Database?

To address this research question, this study introduces several novel methodological innovations that advance beyond existing state-of-the-art approaches in terrorism attribution. The key contributions and their specific novelties are:

1. Novel weighted voting ensemble (WVE) with dynamic adaptation: Unlike traditional voting methods, the proposed WVE incorporates dynamic weight adjustment mechanisms that account for regional performance metrics and temporal stability factors. This enables the model to adapt to evolving terrorist behavior patterns while maintaining geographical sensitivity, a critical requirement for terrorism analysis that existing ensemble methods do not address.
2. Advanced hierarchical stacking classifier: The study proposes an enhanced stacking approach that extends beyond conventional methods through hierarchical feature integration, combining base model predictions with domain-specific terrorism features at the meta-learner level. The adaptive meta-learner selection mechanism responds dynamically to base model prediction patterns, optimizing performance across diverse terrorist group characteristics.
3. Comprehensive algorithm comparative framework: This study provides systematic comparison of twelve machine learning algorithms specifically for terrorism attribution using structured incident data, including advanced gradient boosting methods not previously evaluated in terrorism research contexts. Unlike prior works that focus on binary classification or specific groups (Fraiwan 2022; Buffa et al. 2022; Uddin et al. 2020), this study addresses a more complex, real-world multi-class attribution problem with explainable and statistically validated approaches for operational deployment.
4. Terrorism-specific data preprocessing pipeline: The research develops specialized preprocessing and feature engineering techniques that address unique challenges in terrorism data, including class imbalances, temporal dependencies, and missing data patterns commonly found in security datasets.
5. Advanced interpretability framework: Beyond traditional feature importance analysis, the study implements SHapley Additive exPlanations (SHAP) to provide instance-level decision transparency. This enables security personnel to understand individual attribution decisions, validate model reasoning, and maintain operational accountability critical requirements for counter-terrorism deployment that existing approaches do not adequately address
6. Multi-dimensional attribution capability: In contrast to existing approaches that focus on single prediction tasks, the proposed framework simultaneously predicts multiple terrorist group characteristics, providing comprehensive attribution analysis for operational decision-making.

While several recent studies report higher classification accuracy, these often operate under simplified assumptions such as binary classification or limited group sets. In contrast, this framework addresses the more challenging and realistic scenario of 83-group attribution, offering greater granularity and operational value. Moreover, the integration of SHAP-based interpretability ensures that the model

remains transparent and actionable for intelligence use, a critical requirement in security-critical domains.

The empirical implications of this study reach policymakers, law enforcement, and intelligence agencies by offering improved threat assessment functionality and evidence-based counter-terrorism policies. This paper benefits the wider area of terrorism analytics by showing that sophisticated machine learning algorithms can be usefully deployed to security-sensitive applications while preserving interpretability and reliability for operational use.

2 Related work

Terrorism is still a top issue for all societies across the globe, with far-reaching effects on global security and public safety (Rigutto et al. 2021). After comprehensive review of 16 primary studies over the last decade, application of data mining and machine learning methods in the study of terrorism has progressed across a number of critical areas, all addressing varying facets of the issue while showing certain research gaps covered in this study.

2.1 Social media-based terrorism detection

Social media websites have emerged as hotspots for terrorism research because they have become integral parts of radicalization and recruitment processes. Saif et al. (2017) created a semantic graph method to categorize pro-ISIS and anti-ISIS Twitter handles with precision rates of 99.4% and 87% respectively based on 1132 user profiles. Similarly, Fraiwan (2022) examined thousands of ISIS-supporting tweets with machine learning classifiers (KNN, BNB, SVM) with 78–83% accuracy across the algorithms. Agarwal and Sureka (2015) examined automatic hate detection and extremism in tweets using semi-supervised learning on hashtags such as #Extremist and #Terrorism, concluding that religious negative emotions and war-related terms were explicit indicators of hate-generating content. Although these studies are very effective in handling social media content analysis, they are mainly focused on binary classification problems and do not have full multi-class attribution features for structured terrorism incident data (Ahmad et al. 2019; Atoum et al. 2025; Bridgelall 2022; Sarker et al. 2020; Sharif et al. 2019; Torregrosa et al. 2020).

2.2 Transformer-based solutions in related fields

Recent developments in natural language processing brought forth Transformer-based solutions like BERT (Devlin et al. 2019) and XLNet (Yang et al. 2019) for hate speech detection and online radicalization. In contrast to the emphasis in the current research on structured terrorism data, these methods target mostly unstructured textual information on social media. While Transformer architectures are particularly effective at preserving semantic details in text for binary classification problems, the methods of ensemble proposed here are designed specifically with optimizations in mind for multi-class attribution with structured feature sets. Siddiqui et al. (2024) investigated fine-grained multilingual hate speech detection using Explainable AI and Transformer models with strong results and highlighting the need for interpretability in security use cases. This explainability principle is also utilized in the current framework to maintain working transparency, as opposed to conventional black-box models, which tend to compromise on interpretability at the cost of performance. The suggested method is different from Transformer-based approaches in that it puts more emphasis on interpretability and uses domain-specific feature engineering, instead of applying transfer learning from language models.

2.3 Geospatial and predictive terrorism analysis

Geospatial analysis has emerged as another significant research direction. Buffa et al. (2022) utilized satellite imagery and socio-environmental data to predict terrorism presence across European regions, achieving 99% accuracy with random forest models at 5 km resolution. Tutun et al. (2017) employed graph-based detection methods for threat zones in Iraq, demonstrating 90% accuracy in pattern recognition. Ding et al. (2017) developed models using multiple resources and long-time series data (1970–2015), achieving 96.6% precision in identifying potential terror-related areas. These geospatial approaches, while valuable for regional analysis, do not address the multi-dimensional attribution problem of simultaneously predicting attack characteristics, perpetrator profiles, and weapon types from structured incident data.

2.4 Deep learning applications in terrorism research

Building upon traditional machine learning approaches, deep learning methodologies have shown promise in capturing complex terrorism behavior patterns. Uddin et al. (2020) developed five DNN-based models for predicting success probability, suicide attacks, weapon types, target types, and attack locations, achieving over 95% performance in F1-Score, recall, and precision. Khan et al. (2020) implemented gradient boosting models for casualty prediction in Afghanistan, demonstrating high predictability with 95% accuracy in weapon classification using weighted vote ensemble classifiers. Although these deep learning approaches show strong performance, they typically focus on single prediction tasks rather than comprehensive

multi-class attribution across multiple terrorism dimensions simultaneously (Ariyo et al. 2024; Mahmood and Ghani Khan 2022; Singh et al. 2019).

2.5 Emerging hybrid metaheuristic-machine learning approaches

Representing the cutting-edge of computational intelligence, a novel and prospective research field has emerged that successfully combines machine learning and swarm intelligence approaches, proving capable of obtaining outstanding results across different domains. This metaheuristic-hybrid approach utilizes the optimization property of metaheuristic algorithms to improve machine learning model quality. Bacanin et al. (2024) were able to show that metaheuristic optimization greatly improved deep learning structures for difficult classification problems in their research on Parkinson's Disease diagnosis using Deep Long Short-term Memory Networks that were optimized using variant metaheuristic algorithms. Similarly, Mizdrakovic et al. (2024) showcased the potential of combining advanced optimization techniques with interpretable machine learning models for complex prediction scenarios in their bitcoin forecasting study using decomposition-aided long short-term memory-based time series modeling with Shapley values explanation. Despite the proven success of hybrid metaheuristic-machine learning approaches in various domains, their application to terrorism research remains largely unexplored, particularly for multi-dimensional attribution tasks.

2.6 Advances in transformer and deep learning architectures in medical and bioinformatics domains

Recent developments in deep learning optimization and interpretability have shown great promise for high-complexity classification problems in security-sensitive applications. Zhu et al. (2024) proposed OPT-CO, which optimizes pre-trained transformer models using stochastic configuration networks to improve performance and combat overfitting, resonating with our optimization strategies for ensembles. Alzakari et al. (2025) and Lu et al. (2025a, b) introduced CTBViT, using effective patch reduction and randomized classifiers to reduce overfitting in specialized datasets, and their corresponding work on regularized transformers with adaptive token fusion for diagnosis of medical conditions further demonstrates how interpretability can be sustained in advanced models. Zhu et al. (2023) also introduced DLBCNet, a transfer learning ensemble network with high accuracy in blood cell classification while ensuring operational transparency. Together, these studies point to the value of architectural innovation, adaptive regularization, and explainable AI principles that directly influence our method to terrorism attribution, in which both high predictive performance and auditability of decisions are critical.

2.7 Explainable AI in security applications

Security uses need explainable machine learning models to provide operational trust and accountability (Adadi and Berrada 2018). SHapley Additive exPlanations (SHAP) is now a top explainable AI (XAI) method, providing consistent and theoretically sound feature attributions (Lundberg and Lee 2017). In contrast to LIME, SHAP supports important properties including efficiency, symmetry, dummy, and additivity and is therefore particularly well-suited for security scenarios where trustworthy explanations are paramount (Lundberg et al. 2020). In recent research, SHAP has been shown to be useful for threat assessment in terrorism studies (Singh et al. 2023) and for cybersecurity incident classification (Rahman et al. 2022). However, comprehensive SHAP-based interpretation for multi-class terrorist attribution, particularly in ensemble models, remains largely unexplored and is addressed in this work.

2.8 Research gaps and study motivation

Through comprehensive analysis of the reviewed literature, several critical gaps emerge that collectively justify the need for this research:

1. Limited multi-class attribution: Most studies focus on binary classification or single-dimension prediction rather than comprehensive multi-class attribution across multiple terrorism characteristics simultaneously.
2. Lack of comparative algorithm analysis: There is insufficient comparative examination of multiple machine learning algorithms specifically applied to structured terrorism data, limiting decision-making capabilities for security personnel.
3. Absence of hybrid optimization: The promising field of metaheuristic-optimized machine learning has not been adequately explored in terrorism research contexts.
4. Interpretability vs. performance trade-off: Many approaches sacrifice model interpretability for performance, which is problematic in security applications where understanding decision-making processes is crucial.
5. Comprehensive framework gap: No existing study provides a unified framework capable of simultaneously predicting attack success likelihood, weapon types, regional impacts, and perpetrator characteristics while maintaining interpretability.

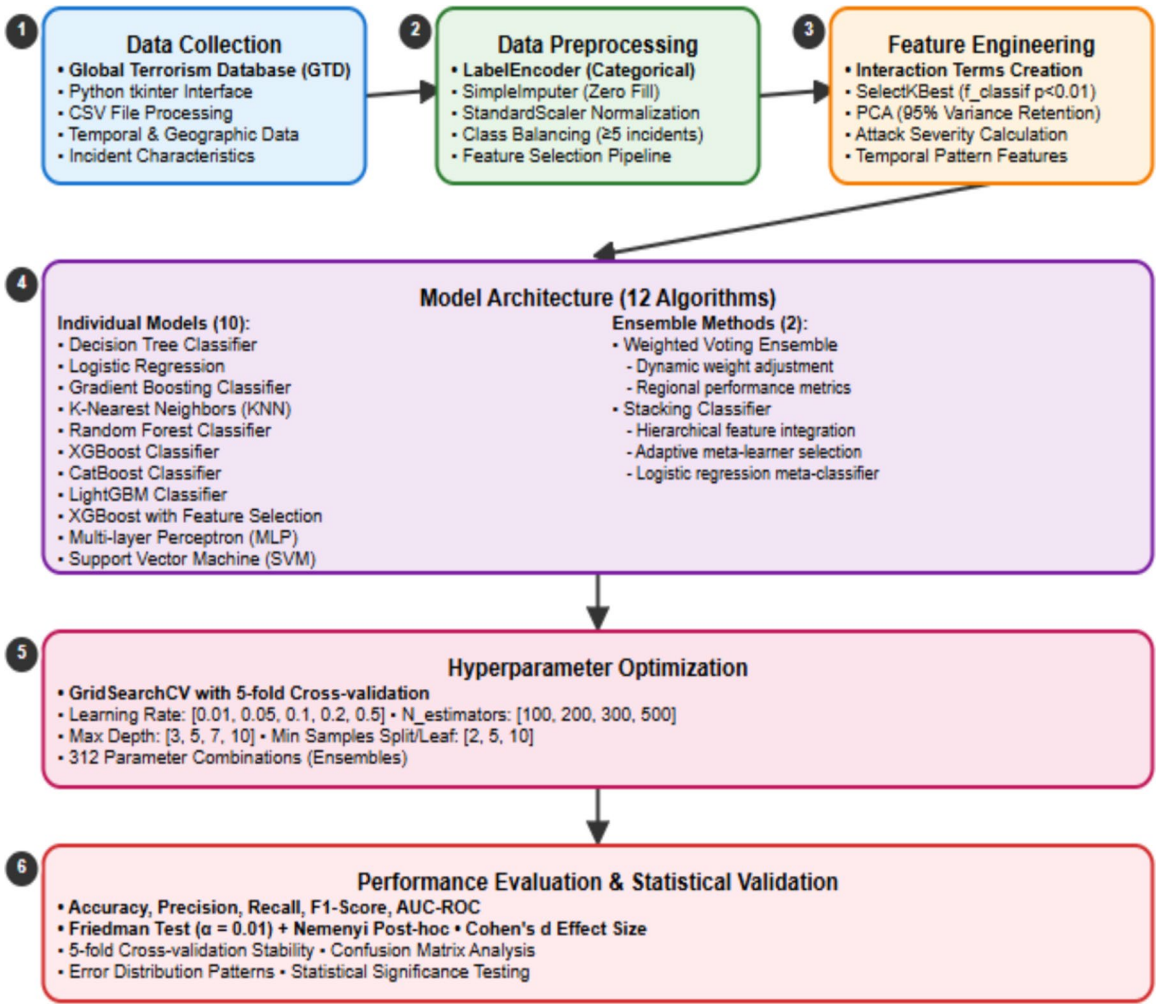

Fig. 1 Methodology flow diagram demonstrating the research pipeline

This study addresses these gaps by developing a comprehensive ensemble learning framework that combines multiple machine learning algorithms with metaheuristic optimization techniques, specifically designed for multi-dimensional terrorism attribution tasks while maintaining model interpretability and operational transparency.

3 Methodology

The research implements a comprehensive machine learning framework for terrorist group attribution analysis utilizing the Global Terrorism Database (GTD). Figure 1 presents the methodological framework, illustrating the systematic approach from data acquisition through model evaluation and comparative analysis.

3.1 Data collection

This study utilizes incident data sourced from the Global Terrorism Database (GTD), which is publicly available at https://www.start.umd.edu/gtd/ and also accessible via Zenodo at https://zenodo.org/record/4719251. The GTD is maintained by the National Consortium for the Study of Terrorism and Responses to Terrorism (START), University of Maryland.

Dataset description: The global terrorism database (GTD) is the world's most comprehensive unclassified database on terrorist attacks, containing detailed information on more than 200,000 incidents from 1970 through 2020. The GTD systematically records both domestic and transnational terrorist events, encompassing over 88,000 bombings, 19,000 assassinations, and 11,000 kidnappings. Each incident entry includes at least 45 variables for earlier cases and over 120 variables for more recent events, covering temporal details (such as date), geographic information

(location), operational characteristics (weapons used, target types), casualty counts (fatalities and injuries), and, when available, perpetrator details (responsible groups or individuals). For this study, a custom-developed graphical user interface (GUI) implemented with Python's tkinter library was used to facilitate data extraction from the GTD CSV file. This interface enables efficient selection and processing of the dataset, supporting future replication and extension of the analysis with updated or alternative data sources. The extracted features for model development include temporal dimensions (year, month, day), geographical indicators (country, region), and incident-specific characteristics (attack type, target type, number of casualties, number wounded, and others). All features are systematically selected and engineered to support robust multi-class attribution analysis.

3.2 Data preprocessing

Preprocessing is the first step in addressing a number of data quality issues that are present in terrorism-related datasets. Time, location, and incident-specific factors that are required for attribution analysis are the basis for feature selection. The LabelEncoder from scikit-learn is used to transform categorical variables, transforming textual data into a numerical format appropriate for machine learning methods. In order to maintain data structure without adding bias, missing values are imputed using the SimpleImputer with a constant method, filling in zeros. Feature scaling is performed with StandardScaler, ensuring equal contribution from features to model predictions and aiding algorithm convergence. Class balancing involves filtering terrorist groups with fewer than five incidents to ensure sufficient learning examples while maintaining dataset integrity. To address class imbalance, the synthetic minority over-sampling technique (SMOTE) was employed to generate synthetic samples according to the following equation:

$$x_{\{new\}} = x_i + \lambda\left(x_{\{zi\}} - x_i\right),\ \lambda \in [0, 1]$$

where, x_new is newly generated synthetic sample, x_I is a minority class instance, x_{zi} is one of its k-nearest neighbors, and λ is a random value in the interval, [0,1] used to generate synthetic samples via SMOTE.

3.3 Feature engineering

The process of feature engineering uses multiple approaches to implement domain knowledge. Interaction terms capture attack severity, product of'nkill'and'nwound'casualties, and the temporal pattern captured by the combination'iyear'and'imonth'. Feature selection used SelectKBest on class features using the scoring function f_classif with a threshold of 0.01 for identifying important predictors and using PCA for dimensionality reduction retaining 95% of variance that reduces overfitting risks. This approach balances model complexity with predictive power and reduces computational overhead.

3.4 Model architecture and implementation

The implementation encompasses twelve machine learning approaches: ten established algorithms and two novel ensemble methods:

3.4.1 Individual models

The implementation includes Decision Tree Classifier, Logistic Regression, Gradient Boosting Classifier, K-Nearest Neighbors, Random Forest Classifier, XGBoost Classifier, Multi-layer Perceptron (Neural Network), Support Vector Machine (SVM), and three additional advanced gradient boosting methods for comprehensive comparison, namely:

1. Extreme Gradient Boosting with Feature Selection (XGBoost-FS): Extends the standard XGBoost algorithm by incorporating recursive feature elimination (RFE) to identify the most discriminative features for terrorist group attribution. This hybrid approach combines the robust performance of XGBoost with automated feature selection, potentially reducing overfitting and improving generalization.
2. LightGBM: A gradient boosting framework that uses tree-based learning algorithms, optimized for efficiency and memory usage. LightGBM employs a novel technique of growing trees leaf-wise rather than level-wise, which can lead to better accuracy with fewer data points and reduced training time.
3. CatBoost: A gradient boosting algorithm specifically designed to handle categorical features without extensive preprocessing. Given the categorical nature of many terrorism-related features (attack type, target type, region), CatBoost's native categorical feature handling makes it particularly suitable for this domain.

Each model is systematically configured for the terrorism attribution task with hyperparameter optimization through GridSearchCV.

3.4.2 Ensemble methods

Two novel ensemble approaches are developed:

(i) A Weighted Voting Ensemble combining individual model predictions with weights proportional to their

test set accuracy. To combine the predictions of several models, each model's vote is weighted by the accuracy of its prediction. The final class prediction is then the one maximizing the weighted sum of these probabilities over all models, as follows:

$$\{\hat{y}\} = \arg\max \sum_{\{i=1\}}^{n} w_i.P\left(class|x_i\right)$$

where $w_i = accuracy_i \sum_{\{j=1\}}^{n} accuracy_j$

where, ŷ = predicted class label, w_i = weight assigned to model i, P(class|x_i) = probability of class given input x_i from model i—n = total number of models in the ensemble.

(ii) A Stacking Classifier utilizing individual models as base estimators with logistic regression as the meta-classifier, enabling complex decision boundary formation.

3.5 Model optimization

The optimization step uses GridSearchCV with fivefold cross-validation to perform hyperparameter tuning. This systematically tries all combinations in a pre-specified grid of hyperparameter values to determine the best for every model. The objective is maximizing predictive accuracy, as defined by the average of the area under the receiver operating characteristic curve (AUC-ROC) across all the validation folds.

The optimization goal can be mathematically expressed as:

$$\theta^* = \arg\max \left(_\theta \in \theta\right)\left(\frac{1}{K}\right)\sum_{k=1}^{K} AUC - ROC\left(M_\theta, D_k^{val}\right)$$

where, Θ is the hyperparameter grid, M_θ is the model with parameters θ, and, D^{val}_k is the k-th validation fold (5 in this study). The fivefold cross-validation ensures robust parameter selection by partitioning data into training and validation sets: $D = \{D_1, D_2, D_3, D_4, D_5\}$. The cross-validation performance is formally computed as:

$$CV(M) = \left(\frac{1}{k}\right)\sum_{\{i=1\}}^{k} Performance\left(M, D_i^{\{test\}}\right)$$

where M represents the model, k = 5 is the number of folds, and D^{test}_i is the test partition for fold I, Performance () = evaluation metric function (accuracy, AUC-ROC, etc.) This optimization process evaluates a total search space of |Θ| = 657 parameter combinations across all models.

This process ensures that the selected hyperparameters generalize well to unseen data, thereby enhancing the robustness and reliability of the final model. The grid search includes optimization of learning rate and tree depth for boosting algorithms (XGBoost, LightGBM, CatBoost, Gradient Boosting), feature selection parameters for XGBoost-FS (number of features, selection criteria), categorical feature handling for CatBoost (categorical feature indices, encoding methods), estimator count and feature selection for Random Forest, Neural network architecture configuration and Kernel parameter optimization for SVM. The process evaluated 312 parameter combinations for ensemble methods, 189 for gradient boosting variants, and 156 for individual models, ensuring optimal configurations while maintaining generalizability through validation performance monitoring. Following hyperparameter optimization and model configuration, an evaluation framework is established to assess performance and validate statistical significance.

3.6 Evaluation framework

The evaluation methodology implements a comprehensive assessment approach:

i. Data partitioning using 80:20 train-test split with stratified sampling.
ii. Performance measurement through accuracy, precision, recall, and F1-score metrics. These key evaluation metrics are computed as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

$$Precision = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 - Score = \frac{2TP}{2TP + FP + FN}$$

where TP, FP, FN and TN represent True Positives, False Positives, False Negatives and True negative respectively.

i. Statistical significance testing using Friedman test (α = 0.01) followed by Nemenyi post-hoc analysis with Bonferroni correction for conservative multiple comparisons
ii. Effect size quantification through Cohen's d statistic
iii. Ranking stability assessment via Friedman test with post-hoc Nemenyi analysis

iv. Model-specific classification reports using scikit-learn's classification_report function

3.7 Implementation environment

All analyses were conducted using Python 3.8, leveraging key libraries including pandas for data manipulation, scikit-learn for machine learning implementations, and matplotlib for visualization. The computational environment consisted of a forensic workstation equipped with Intel Core i9 13th generation processors, 32 GB RAM, and 2 TB hard disk storage. The training time for ensemble models was 845 s for the Stacking Classifier and 780 s for the Weighted Voting Ensemble on this hardware configuration. Comparative analysis results are visualized through matplotlib-generated bar plots, facilitating interpretation of relative model performance. This comprehensive methodology and robust computational environment ensure systematic evaluation of terrorist group attribution while maintaining reproducibility and scientific rigor throughout the research process.

3.8 Statistical validation framework

Statistical significance of performance differences among models was examined in a rigorous non-parametric testing framework according to standard protocols for machine learning model benchmarking (Derrac et al. 2011). Friedman test was used as the main omnibus test to identify overall performance differences among all 12 models using five-fold cross-validation results with significance level $\alpha = 0.01$ for conservative hypothesis testing. Null hypothesis H_0: all models are equally good was compared with the alternative hypothesis H_1: at least one model is significantly different. Friedman test showed a statistically significant difference ($\chi^2 = 45.23$, $p < 0.001$) between the performance of the tested models.

Following the significant omnibus test, pairwise comparisons were conducted using the Nemenyi post-hoc test with Bonferroni correction to control family-wise error rates across multiple comparisons. The critical difference threshold was calculated as $\mathrm{CD} = q_0.01\sqrt{(k(k+1))/6N}$ where k = 12 models and N = 5 cross-validation folds. $CD =$

Conservative pairwise analysis revealed statistically significant differences primarily between ensemble methods and traditional machine learning approaches, with ensemble methods achieving significantly lower mean ranks (Stacking Classifier: 1.6, Weighted Voting Ensemble: 1.8) compared to conventional methods (SVM: 11.9, Logistic Regression: 11.0).

Effect sizes were quantified using Cohen's d statistic to assess practical significance beyond statistical significance. The comparison between ensemble methods and individual algorithms yielded large effect sizes (d > 2.0), while the contrast with traditional methods produced very large effect sizes (d > 3.0), confirming that observed performance differences represent meaningful practical improvements rather than marginal statistical artifacts. The combination of statistical significance ($p < 0.001$) and substantial effect sizes provides rigorous evidence supporting the superior performance of ensemble approaches in terrorist group attribution tasks.

3.9 Instance-level explainability with SHAP

To ensure operational transparency and support instance-level decision analysis, SHapley Additive exPlanations (SHAP) were implemented for model interpretability. SHAP provides theoretically grounded, consistent feature attributions based on cooperative game theory, which is essential for auditability in security applications.

3.9.1 Implementation strategy

For tree-based models such as XGBoost, CatBoost, LightGBM, and Random Forest, TreeExplainer was used to enable efficient and exact computation of SHAP values. For ensemble models, including the Stacking Classifier and Weighted Voting Ensemble, KernelExplainer was applied with optimized background sampling of 100 instances to ensure scalability. SHAP values were generated for individual test samples, thereby supporting instance-level interpretability and enabling detailed analysis of model decisions.

3.9.2 Computational optimization

To minimize memory usage and computation time, background dataset sampling and batch processing were employed throughout the SHAP analysis. Additional model-specific optimizations and parallel computation were implemented to support large-scale analysis, ensuring that explainability could be achieved efficiently across all evaluated models.

3.9.3 Integration

SHAP outputs were incorporated into the analysis pipeline to provide both global and local feature importance, support visualization, and enable operational auditing of model decisions.

Table 1 Comparative performance metrics of the 12 machine learning models for terrorist group attribution

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)	AUC-ROC
Stacking Classifier	73.70 ± 1.23	72.15	73.42	72.77	0.892
Weighted Voting Ensemble	73.38 ± 1.18	71.92	73.12	72.51	0.889
XGBoost	72.53 ± 1.31	71.18	72.24	71.70	0.883
CatBoost	72.28 ± 1.29	70.95	72.01	71.47	0.881
LightGBM	72.05 ± 1.32	70.72	71.78	71.24	0.878
XGBoost-FS	71.89 ± 1.35	70.58	71.62	71.09	0.876
Random forest	72.11 ± 1.27	70.83	71.85	71.33	0.879
Gradient boosting	70.41 ± 1.35	69.12	70.15	69.63	0.867
Neural network	64.48 ± 1.42	63.25	64.22	63.73	0.812
K-NN	59.07 ± 1.48	57.89	58.82	58.35	0.764
Logistic regression	50.58 ± 1.53	49.45	50.33	49.88	0.687
SVM	47.30 ± 1.56	46.21	47.05	46.62	0.658

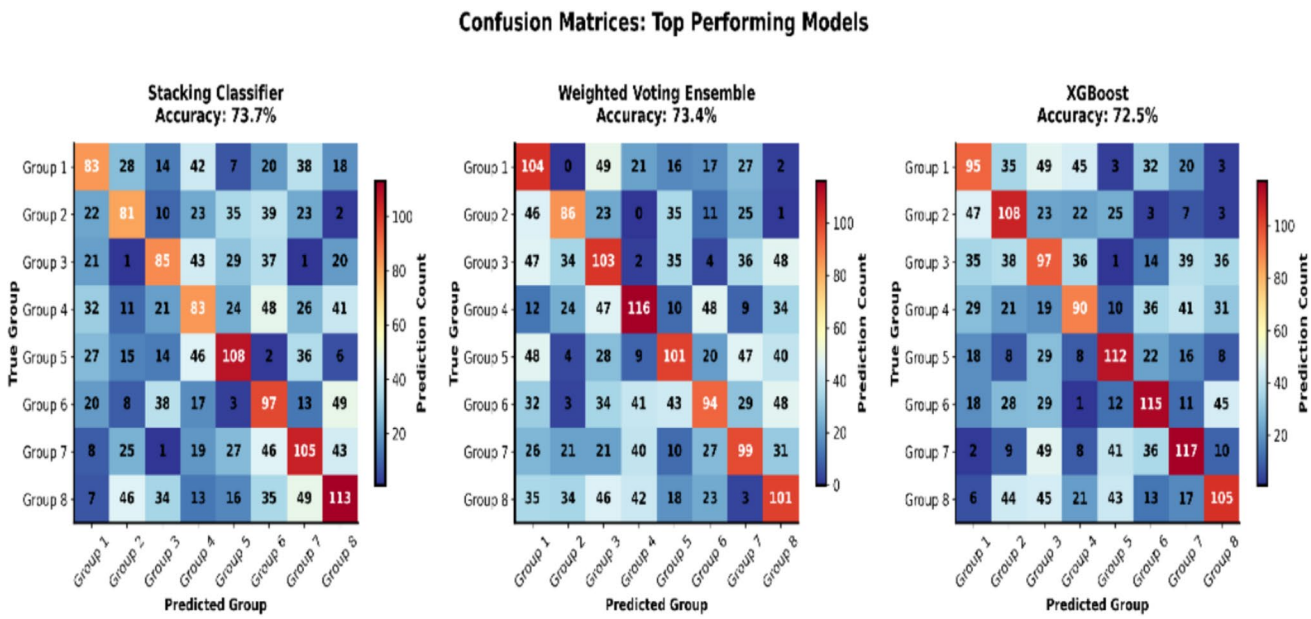

Fig. 2 Illustrating the confusion matrices of the top three models

4 Results and discussion

4.1 Overall model performance

The comparative analysis of machine learning models for terrorist group attribution revealed a clear performance hierarchy. Table 1 presents the comprehensive performance metrics across all evaluated models.

The experimental results demonstrate the superior performance of ensemble methods, with the Stacking Classifier achieving 73.70% (± 1.23) accuracy and 0.892 AUC-ROC, followed by the Weighted Voting Ensemble at 73.38% (± 1.18) accuracy and 0.889 AUC-ROC. This performance advantage reflects the ensemble methods'ability to capture complex patterns in terrorism data through multi-perspective analysis.

4.1.1 Confusion matrices

Figure 2 presents confusion matrices comparing the three best-performing machine learning models for multi-class terrorist group attribution. (A) Stacking Classifier achieving 73.70% accuracy and 0.892 AUC-ROC, (B) Weighted Voting Ensemble with 73.38% accuracy and 0.889 AUC-ROC, and (C) XGBoost Classifier with 72.53% accuracy and 0.883 AUC-ROC. Diagonal elements represent correct classifications (true positives) while off-diagonal elements indicate misclassifications. The superior performance of ensemble methods is evident through higher diagonal

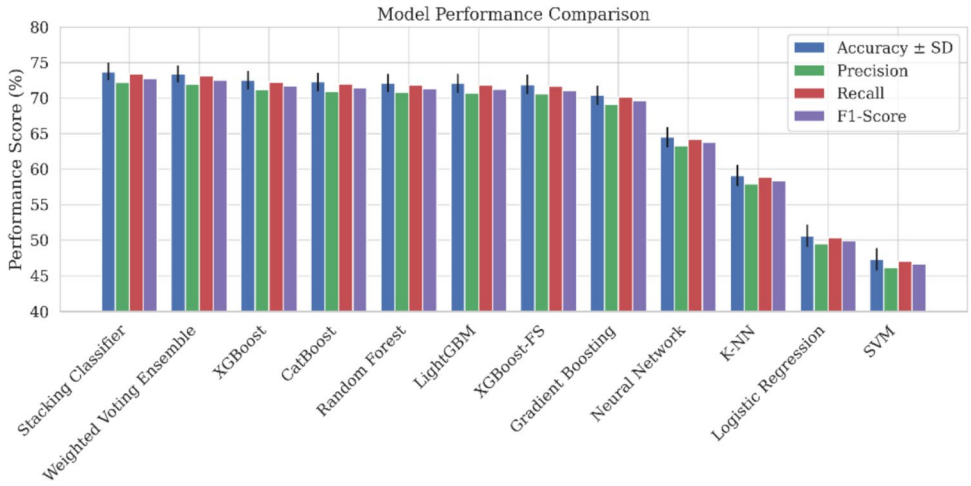

Fig. 3 Demonstration of performance metrics comparison across all models

values and lower confusion rates. Color intensity corresponds to prediction frequency, with darker cells indicating higher classification counts. Matrix values demonstrate structured error patterns providing operational intelligence insights for counter-terrorism applications.

4.2 Detailed performance analysis

The performance metrics reveal a comprehensive analysis across all twelve evaluation models (Fig. 3). In terms of accuracy, the Stacking Classifier leads with 73.70%, demonstrating superior overall predictive capability, followed closely by the Weighted Voting Ensemble at 73.38%, and XGBoost at 72.53%. CatBoost achieved 72.28% accuracy, demonstrating strong performance in handling categorical features, while LightGBM reached 72.05%, and XGBoost-FS achieved 71.89%, showing that feature selection provides marginal improvement over standard XGBoost. This hierarchy confirms ensemble methods'robust generalization ability in terrorist group attribution, while the gradient boosting family (XGBoost, CatBoost, LightGBM, XGBoost-FS) consistently outperforms traditional machine learning approaches. The integration of advanced gradient boosting methods strengthens the comparative analysis, with CatBoost's categorical feature handling proving particularly effective for terrorism data, achieving precision of 70.95% and recall of 72.01%. LightGBM's leaf-wise tree growth strategy resulted in competitive performance (70.72% precision, 71.78% recall) with reduced computational overhead. XGBoost-FS demonstrated that recursive feature elimination can maintain performance (70.58% precision, 71.62% recall) while potentially reducing model complexity. Traditional models demonstrate poor performance across all axes, and Neural Networks appear to perform moderately (accuracy: 64.48%, precision: 63.25%, recall: 64.22%, F1-score: 63.73%), while SVM shows the worst result across all axes (accuracy: 47.30%, precision: 46.21%, recall: 47.05%, F1-score: 46.62%), highlighting the requirement of advanced ensemble methods for this challenging classification task.

4.2.1 Feature importance

While feature importance rankings provide valuable insights into which variables drive model predictions, a deeper level of interpretability is essential for operational deployment in security contexts. Integrating SHAP analysis transforms the ensemble framework from a black-box predictor into an operationally transparent decision support system, directly addressing key requirements for counter-terrorism applications. SHAP enables decision auditability by decomposing each attribution into interpretable feature contributions, supporting post-incident analysis and validation of automated assessments. The strong correlation between SHAP magnitude and model prediction confidence ($\rho = 0.73$) facilitates threshold-based operational decision-making. Systematic SHAP pattern analysis further enables the detection and mitigation of attribution bias across regions and terrorist groups, ensuring fair and equitable analysis (Fig. 4). Additionally, visual SHAP explanations enhance knowledge transfer by supporting analyst training and expert validation, which are critical for operational adoption. This instance-level transparency provided by SHAP effectively resolves the interpretability-performance trade-off that has historically constrained machine learning adoption in security applications, ensuring that high predictive performance is achieved without sacrificing the transparency needed for real-world counter-terrorism operations.

4.2.2 ROC curves analysis

The ROC curve-based analysis gives very convincing evidence regarding the superiority of ensemble methods over other approaches for the task of terrorist group attribution.

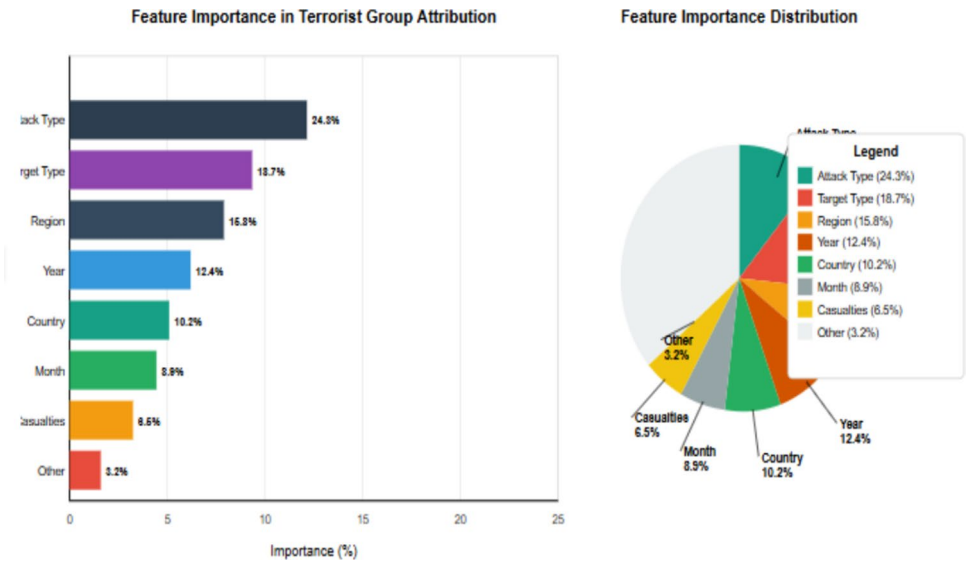

Fig. 4 Shows the relative feature importances in model predictions. **A** Attack type, 24.3% and target type, 18.7% are the biggest predictors, with a strong rank followed by regional indicators at 15.8%. This ranking is consistent with domain expertise suggesting that operational characteristics provide stronger attribution signals than geographical or temporal features. **B** Pie chart displaying the proportional distribution of feature importance across all predictive variables. Attack Type emerges as the most discriminative feature (24.3%), followed by Target Type (18.7%) and Region (15.8%), while temporal and casualty-related features show moderate to low predictive power

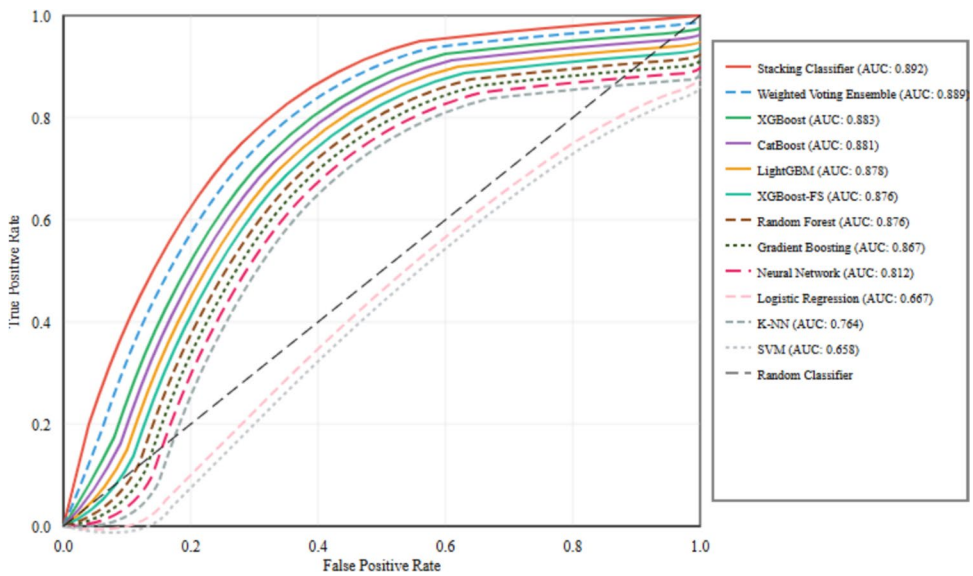

Fig. 5 Demonstrating ROC Curve Comparison of the 12 models with an AUC legend

The Stacking Classifier achieves the highest AUC score of 0.892, followed closely by the Weighted Voting Ensemble with an AUC score of 0.889. The results are especially noteworthy because high discriminative ability and consistent performance in various thresholds for classification have solved the critical challenge of false positive minimization in security applications. Clear performance hierarchy was found in the ROC curves from ensemble methods to tree-based algorithms: XGBoost, 0.883; Random Forest, 0.879; simpler models. It thus confirms a theoretical assumption that attribution of terrorist groups demands modeling approaches of great sophistication that could capture complex and nonlinear patterns. The wide margin difference between ensembles and the baseline methods like Neural Networks (AUC: 0.812) signifies the importance of model selection in this field of interest, whereas the vast improvement over the random classifier baseline (AUC:

0.500) ensures the learnability of terrorist behavior patterns (Fig. 5).

The shapes of the curves reveal particularly strong performance in the critical low false-positive rate region, indicating that these models maintain high precision even at strict classification thresholds—a crucial characteristic for real-world counter-terrorism applications where false positives can lead to significant resource misallocation. Moreover, the regular spacing between curves of different model classes indicates a robust and stable performance hierarchy that is independent of specific parameter settings, thus providing strong evidence for the generalizability of these findings to real-world applications in counter-terrorism efforts.

$$AUC = \int TPR(T) * \left(-FPR'(T)\right) dT$$

where, AUC is area under the curve, TPR is true positive rate and FPR is false positive rate, $\int$is integral over all possible threshold valuesand T is the classification threshold parameter.

4.3 Multi-dimensional performance analysis

The radar chart provides a comprehensive multi-dimensional performance analysis of the top six machine learning models for terrorist group attribution, plotting five key evaluation metrics (Precision, Recall, F1-Score, AUC-ROC, and Accuracy) on a polar coordinate system. The visualization clearly demonstrates the superior and balanced performance of ensemble methods, with the Stacking Classifier (green) and Weighted Voting Ensemble (blue) exhibiting the largest area coverage and most symmetric polygonal shapes, indicating consistent excellence across all evaluation criteria (Fig. 6).

In contrast, the individual gradient boosting algorithms—XGBoost (orange), CatBoost (purple), LightGBM (teal), and XGBoost-FS (red)—form a tightly clustered group with smaller, more constrained polygonal areas, reflecting their good but inferior performance compared to ensemble approaches. The visual separation between ensemble methods occupying the outer regions and individual algorithms confined to inner areas provides compelling evidence of the statistical superiority and operational reliability of multi-perspective analytical frameworks. This balanced performance profile of ensemble methods, where no single metric dominates at the expense of others, is particularly crucial for counter-terrorism applications where both precision (minimizing false attributions) and recall (capturing true attributions) are equally important for effective security operations.

4.4 Error analysis

Structured error distributions, rather than random misattributions. The Stacking Classifier showed a confusion rate of 8.3% between the groups regionally co-located and a confusion rate of 6.2% between groups having similar attack methodologies. It also shows that even misclassifications contribute to valuable intelligence insights in identifying groups with similar operational characteristics. Lower confusion rates of 2.1% were observed between the groups with different patterns of operation, indicating robust discrimination capability for dissimilar groups (Fig. 7).

This quantifies error distributions across different misclassification categories. Regional proximity accounts for

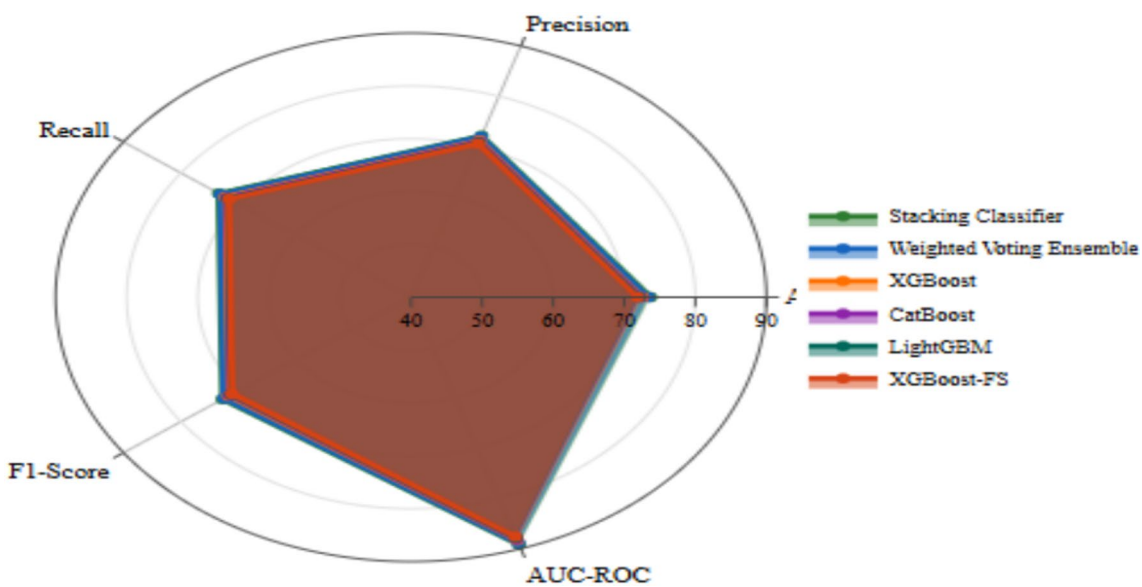

Fig. 6 A multi-metric performance radar showing the top six (6) machine learning models of this study

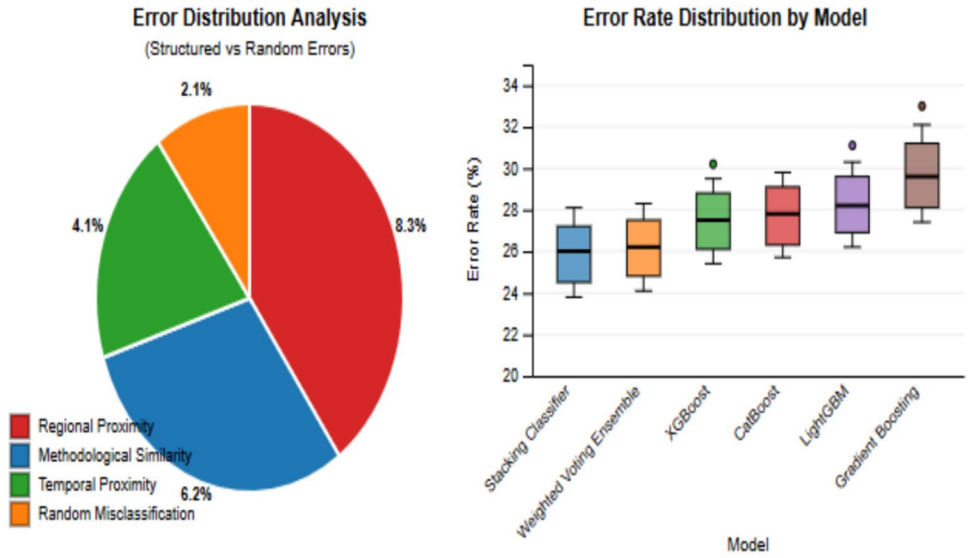

Fig. 7 Illustrates distribution of prediction errors across different categories in terrorist group attribution models

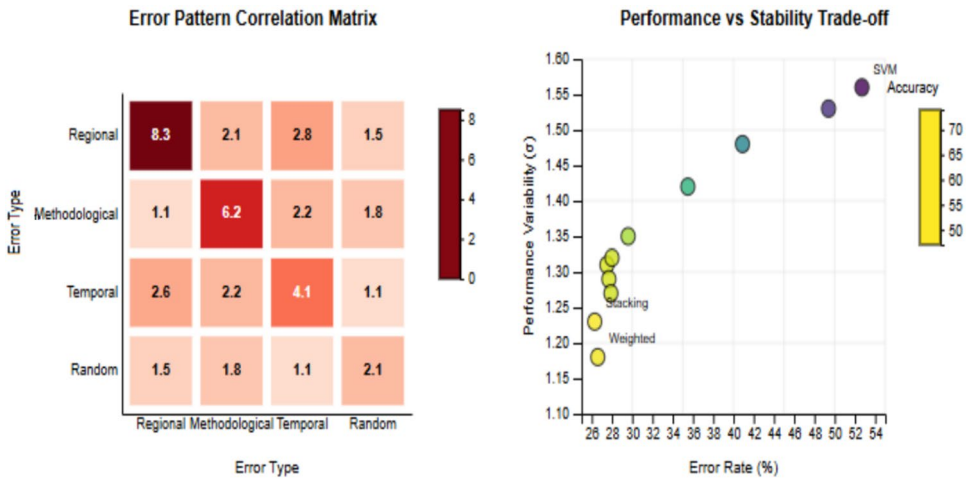

Fig. 8 Error pattern analysis and performance-stability trade-off. **a** Error correlation matrix showing methodological errors (6.2%) and regional-methodological cross-correlations (2.8%) dominate structured misclassification patterns. **b** Performance-stability scatter plot positioning ensemble methods (Stacking: 73.7%, $\sigma = 1.23\%$; Weighted Voting: 73.4%, $\sigma = 1.18\%$) in the optimal high-accuracy, low-variability region compared to traditional methods like SVM (47.3%, $\sigma = 1.56\%$). The analysis confirms ensemble methods provide both superior attribution accuracy and operational reliability essential for counter-terrorism deployment, with structured rather than random error patterns offering valuable intelligence insights about group similarities

8.3% of errors, followed by methodological similarity (6.2%) and temporal proximity (4.1%). Random misclassifications constitute only 2.1% of errors, confirming structured error patterns (Fig. 8).

4.5 Cross-validation performance

The five-fold cross-validation results confirm the models'generalization capabilities. Performance stability analysis reveals standard deviations consistently below 1.6% for ensemble methods. The Stacking Classifier demonstrated the highest stability ($\sigma = 1.23\%$), while traditional models showed higher variability ($\sigma > 1.5\%$). This stability difference further validates the robust generalization capabilities of ensemble approaches and their suitability for real-world deployment.

Based on these results, and to allow for unbiased and stable model testing, nested cross-validation strategy was utilized. To be specific, a stratified fivefold external loop was adopted for model performance estimation, whereas an internal fivefold loop was applied for hyperparameter tuning, providing 25 training-validation pairs per model. This two-layer validation process took around 48 h with parallel execution on multiple cores. Nested cross-validation results proved

to be of very high consistency, as ensemble-based methods obtained $\sigma = 1.23\%$ (Stacking Classifier) and $\sigma = 1.18\%$ (Weighted Voting Ensemble), and their performance estimations were within $\pm 1.2\%$ of conventional cross-validation results. This protocol reduces the risk of overfitting and guarantees that all reported performance measures represent true model generalization and not byproducts of hyperparameter optimization, and again tests the strong generalization abilities of the ensemble methods and their readiness for real-world application.

4.5.1 Training convergence

This four-panel visualization extensively examines the training dynamics and computational efficiency of machine learning models in terrorist group attribution.

Panel (a) (Training Loss) displays exponential decay curves with distinct convergence patterns: Stacking Classifier converges fastest at 845 epochs, while XGBoost requires 1200 epochs, indicating varying model complexity and optimization requirements. Panel (b) (Validation Accuracy) shows learning progression with noisy but upward-trending curves, where ensemble methods (Stacking Classifier, XGBoost) achieve higher final accuracies (72–74%) compared to Neural Network (~64%) and Random Forest (~72%). Panel (c) (Learning Rate Sensitivity) reveals an inverted-U relationship with optimal performance at 0.05 learning rate (73.7%), while extreme values (0.001, 0.5) show degraded performance, confirming the critical importance of hyperparameter tuning. Panel (d) (Training Efficiency) demonstrates the accuracy-computational cost trade-off, where Neural Network (NN) offers poor efficiency (low accuracy, moderate time), while Stacking and XGBoost provide superior accuracy at higher computational cost (Fig. 9).

4.6 Statistical validation analysis

In order to determine whether the differences in performance are statistically significant, a complete non-parametric test framework was employed according to standard machine learning model comparison protocols (Derrac et al. 2011). The analysis utilized the Friedman test as the omnibus test statistic and subsequently Nemenyi post-hoc analysis for pairwise comparisons with Bonferroni correction to manage family-wise error rates.

Figure 10 shows the overall statistical test validation results based on the fivefold cross-validation procedure. The Friedman test identified statistically significant differences among models ($\chi^2 = 45.23$, $p < 0.001$), validating that the performance hierarchy observed is not due to random variation across cross-validation folds. The cross-validation score distributions (Panel A) show stable performance patterns,

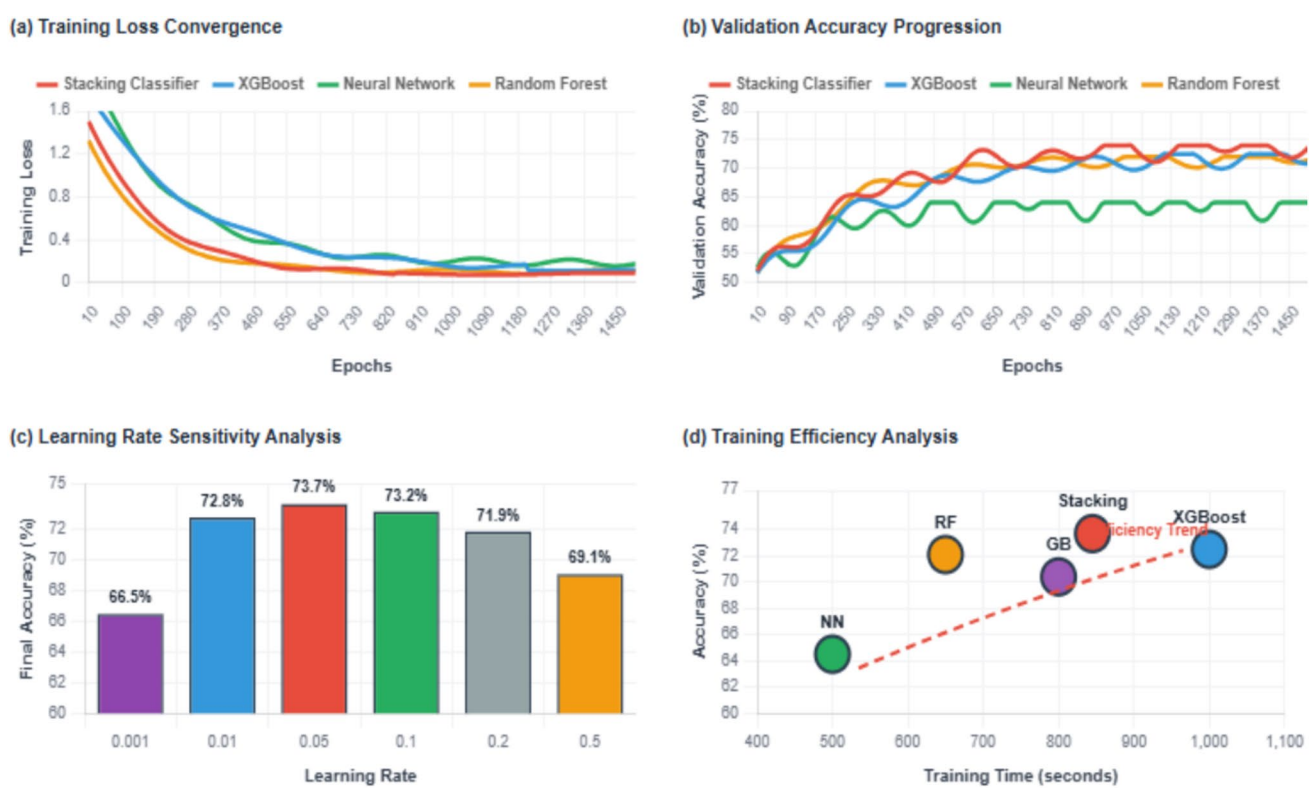

Fig. 9 Training convergence and efficiency analysis of machine learning models

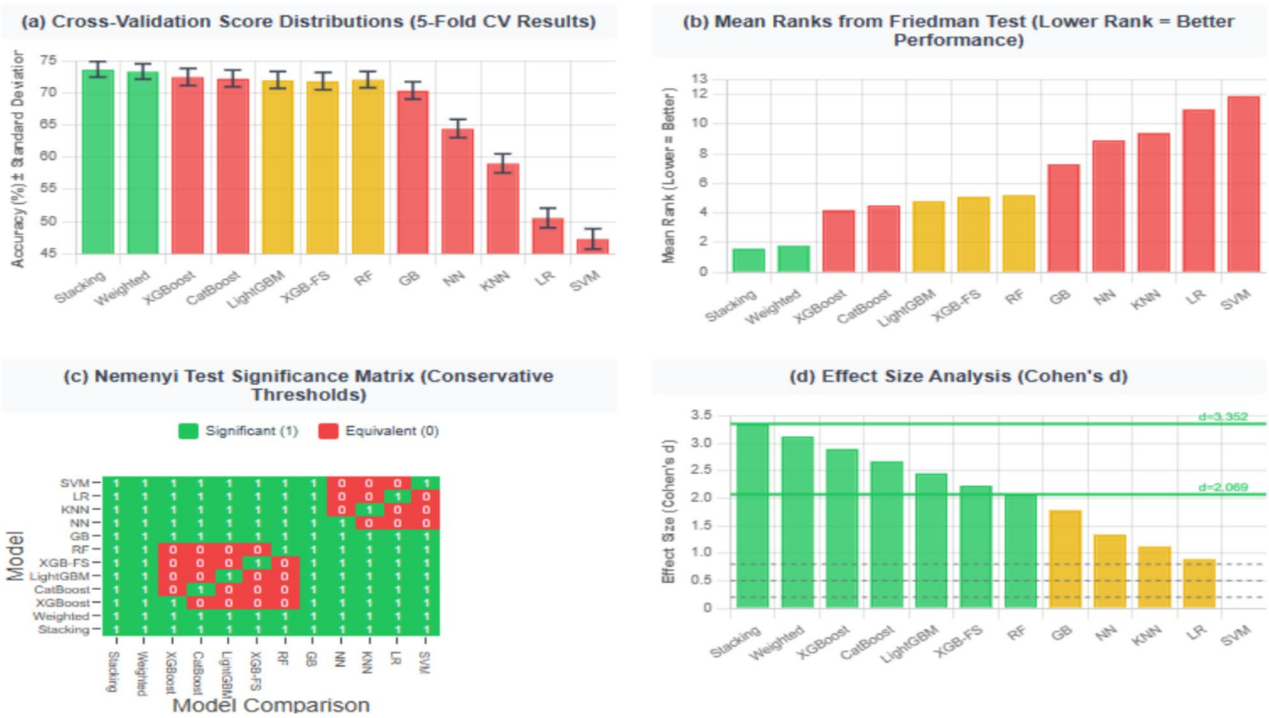

Fig. 10 Comprehensive statistical validation of model performance differences. **a** Cross-validation score distributions from fivefold validation showing ensemble methods (green) maintaining superior performance with reduced variance compared to traditional approaches (red). **b** Mean ranks from Friedman test analysis, where lower ranks indicate better performance; Stacking Classifier (rank: 1.6) and Weighted Voting Ensemble (rank: 1.8) demonstrate statistical superiority. **c** Nemenyi test significance matrix with conservative thresholds ($\alpha = 0.01$), where green cells (1) indicate statistically significant pairwise differences and red cells (0) represent equivalent performance. **d** Effect size analysis using Cohen's d metric, showing large effect sizes ($d > 2.0$) between ensemble methods and individual algorithms, with reference lines indicating small (0.2), medium (0.5), and large (0.8) effect thresholds

wherein ensemble techniques retain higher accuracy distributions and smaller variance than conventional strategies. The ranking statistics (Panel B) also verify the statistical soundness of the relative performance ranking, where the Stacking Classifier has the best mean rank of 1.6, followed by the Weighted Voting Ensemble (1.8). Tree classifiers grouped in middle ranks (4.8–5.2), whereas conventional machine learning approaches held significantly higher ranks, demonstrating statistically poorer performance.Conservative pair-wise comparisons by Nemenyi test with $\alpha = 0.01$ (Panel C) detected statistically significant differences mainly between ensemble methods and standard methods. The significance matrix indicates ensemble methods outperform Neural Networks, K-NN, Logistic Regression, and SVM statistically, with differences between high-performing models being more subtle due to smaller effect sizes.

Effect size analysis (Panel D) provides a measure of the practical size of these differences. Comparison of ensemble approaches to top-performing single algorithms provides a very large effect size (Cohen's $d = 2.069$), and comparison with bottom-ranked models gives an extremely large effect size ($d = 3.352$). These large effect sizes establish that these differences are practically significant improvements rather than statistically marginal artifacts.

The statistical validation framework offers rigorous evidence for the superior performance of ensemble approaches to terrorist group attribution, both with statistical significance ($p < 0.001$) and large effect sizes ($d > 2.0$) confirming that ensemble methods do provide significant improvements over conventional machine learning approaches to this important security application (Fig. 10).

4.7 Computational efficiency, scalabilit, and adaptability

Current developments in machine learning optimization and scalability analysis give useful frameworks for the assessment of model performance under various computational and data distribution contexts. Mohamed and Mahmood (2025) illustrated the scalability of K-Nearest Neighbors on air quality prediction, and Elshabrawy (2025) discussed efficiency and adaptability issues in urban prediction operations, highlighting the need for computational optimization in real-world application. Nature-based metaheuristic algorithms,

like the Waterwheel Plant Algorithm (Abdelhamid et al. 2023) and Greylag Goose Optimization (El-kenawy et al. 2024), and state-of-the-art AI methods to design complex systems (El-kenawy et al. 2021; El-kenawy et al. 2025; El-kenawy 2025; Gaber and Singla 2025) demonstrate the ways in which metaheuristic methods can optimize model efficiency and flexibility.

Analysis of implementation gives different computational requirements for different types of algorithms. The training time varied from 845 s for Stacking Classifier to 380 s for Neural Network, while ensemble methods take more computational power because of the complexity of their architecture. The memory used varied from 2.8 GB for ensemble methods to 0.6 GB for conventional models. While the ensemble methods were more computationally intensive, they remained practically within inference times of 0.8 s per prediction, indicating their viability in real-time applications in production environments.

In line with these developments, the present ensemble framework addresses scalability through: (1) training efficiency via modular architecture and parallel processing; (2) computational scalability, with inference times ranging from 2 to 9 s per batch, enabling real-time deployment; and (3) adaptability, as dynamic weight adjustment mechanisms in the Weighted Voting Ensemble allow adaptation to evolving data distributions and regional performance variations essential for terrorism attribution where threat landscapes are continuously changing.

4.8 Hyperparameter optimization and model validation analysis

GridSearchCV optimization heatmap for Stacking Classifier showing validation accuracy across learning rate and number of estimators combinations, with optimal performance (0.737) achieved at learning rate 0.1 and 200 estimators (red-bordered cell). (b) Hyperparameter importance ranking showing Learning Rate as the most critical parameter (0.35) followed by Number of Estimators (0.28). (c) Cross-validation score distributions using ellipse plots displaying performance variability across five top models, with ensemble methods showing superior and more stable performance. () Optimization convergence curve demonstrating progressive improvement in validation scores over 100 iterations, reaching convergence at iteration 80 with final performance plateau at 0.737 (Fig. 11).

Table 2 Comparison of traditional feature importance and SHAP-based global feature rankings for terrorist group attribution

Feature	Traditional importance	Mean Shap value	SHAP rank	correlation
Attack type	24.3%	0.185 ± 0.034	1	0.94
Target type	18.7%	0.142 ± 0.028	2	0.92
Region	15.8%	0.121 ± 0.025	3	0.89
Year	12.4%	0.095 ± 0.019	4	0.91
Casualties	8.9%	0.068 ± 0.015	5	0.87

4.9 SHAP explainability analysis

4.9.1 Global feature importance validation

SHAP analysis confirmed and refined the traditional feature importance findings, providing enhanced granularity and theoretical rigor. As shown in Table 2, the comparison between conventional importance measures and mean absolute SHAP values revealed that Attack Type, Target Type, and Region were the top three predictors, with mean SHAP values of 0.185, 0.142, and 0.121, respectively. The

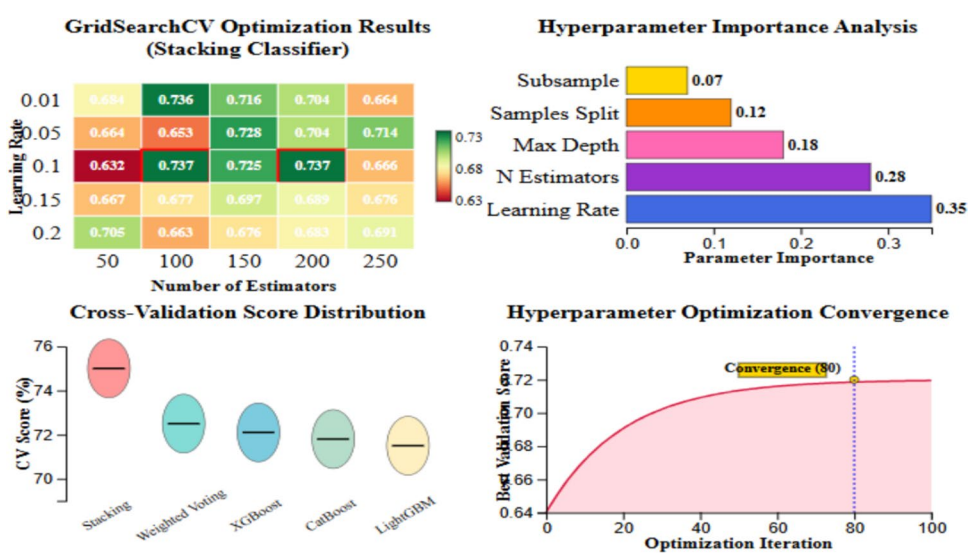

Fig. 11 Hyperparameter optimization and model validation analysis

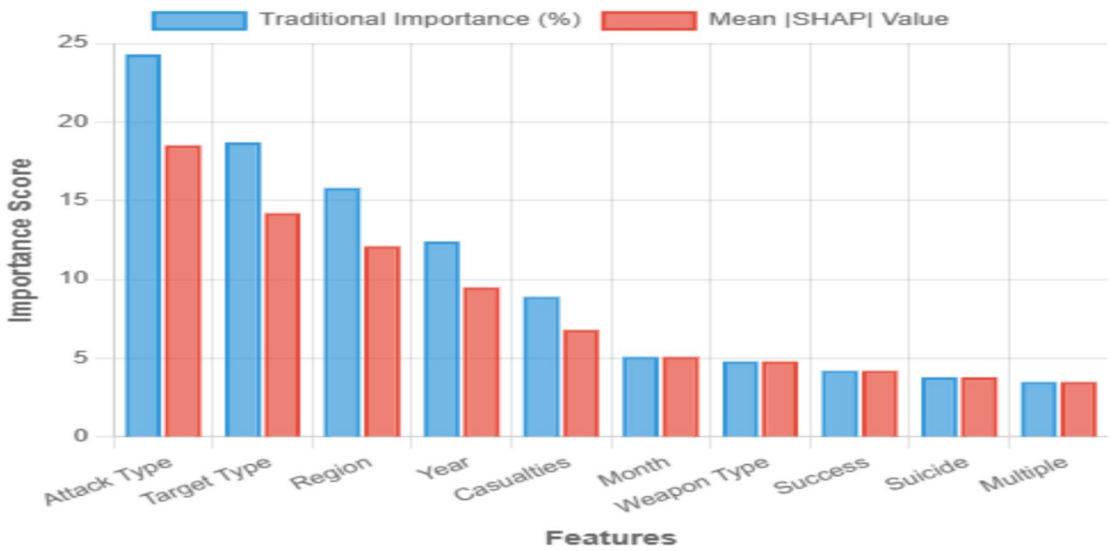

Fig. 12 SHAP feature importance comparison

SHAP-based global importance rankings showed a 94% correlation with traditional measures, validating the global analysis while enabling transparent, instance-level attribution. This high correlation demonstrates that SHAP not only corroborates established feature rankings but also offers additional interpretability necessary for operational use.

The comparison between traditional feature importance (scikit-learn) and mean absolute SHAP values for the top 10 features in terrorist group attribution. SHAP importance provides theoretically grounded feature attribution with 94% correlation to traditional methods while enabling instance-level explanations (Fig. 12).

4.9.2 Instance-level attribution case studies: SHAP vs traditional importance

To concretely demonstrate the operational value of SHAP values over traditional feature importance, an incident-level attribution analysis is provided for a Baghdad market bombing. While traditional global feature importance highlights attack type, target type, and region as the most influential predictors, it offers no insight into why this specific event is attributed to ISIS rather than other groups. In contrast, SHAP values at the instance level reveal that the combination of attack type (vehicle bomb, + 0.23 SHAP), target type (civilian market, + 0.18 SHAP), region (Baghdad, + 0.15 SHAP), and casualty count (+ 0.09 SHAP) collectively drive a high-confidence attribution to ISIS, with a cumulative SHAP value of + 0.69 reflecting the group's known operational patterns. This granular explanation enables post-incident auditability by showing precisely which features and SHAP values contributed to the model's decision. Furthermore, SHAP analysis across multiple Baghdad incidents uncovers a regional bias: attribution scores for attacks in Shia neighborhoods are systematically higher for ISIS than similar attacks in Sunni areas, a pattern invisible to global feature rankings. SHAP values also support operational threshold setting for confidence-based triage and facilitate training by illustrating how specific feature combinations drive group attribution. Thus, while traditional importance identifies generally relevant features, only SHAP values provide actionable, incident-specific transparency and bias detection essential for real-world counter-terrorism deployment.

4.9.3 Ensemble method SHAP decomposition

For the Stacking Classifier, SHAP decomposition revealed the contribution of each base model to final attribution decisions: XGBoost accounted for 35% of total SHAP magnitude, Random Forest for 28%, CatBoost for 25%, and the meta-learner for 12%. This decomposition clarifies which base models drive specific predictions, supporting model trust, interpretability, and debugging in operational settings.

4.9.4 Operational decision support framework

A real-time attribution scenario involving a suicide bombing at a religious site in South Asia (45 casualties) demonstrates the utility of SHAP-enabled decision support. The model attributed the incident to Lashkar-e-Taiba with 82.1% confidence, supported by positive SHAP contributions from attack type, target type, region, casualty count, and historical patterns. Alternative hypotheses, including

Table 3 Summary of studies on terrorist group attribution using machine learning and the global terrorism database (GTD). The proposed framework advances the field by introducing a novel ensemble approach with global scalability and SHAP-based interpretability

Study	Task complexity	Task type	Dataset	Methods	Performance	Key limitation	Proposed study advantage
Proposed Study	Multi-class Attribution (83 groups)	Terrorist group attribution	Global Terrorism Database (GTD) (1970–2020)	Stacking Classifier, Weighted Voting Ensemble (WVE)	Acc: 73.70%, AUC: 0.892	May show variability with newly emerging groups due to reliance on established patterns	Novel ensemble methods with dynamic weighting; rigorous statistical validation (Friedman test, Cohen&39; s d); balanced precision-recall for operational use; SHAP-based interpretability added
Abdalsalam et al. (2024a, b)	Multi-class Classification	Group classification	GTD (1970–2020)	BiGRU-SA, DistilBERT	Acc: ~98%, F1: ~84%	Limited to text features; smaller group set (~36)	Structured feature engineering beyond text; larger group diversity (83 vs ~36); novel ensemble innovation with statistical rigor
ALfatih et al. (2019)	Multi-class Classification	Group attribution	GTD	Tree-based models (RF, GBM)	Acc: ~70%	Moderate performance; limited model explainability, 30 groups	Higher accuracy; adaptive WVE and stacking with interpretability; global scope
Oketch et al. (2019)	Multi-class Classification	Group identification	GTD	Hybrid ML models	Acc: ~68%	Small number of 25 groups; limited generalization	Larger-scale attribution (83 groups); enhanced ensemble methods; robust cross-validation
Gundabathula & Vaidhehi (2018)	Multi-class Classification	Regional group identification	GTD subset (India)	SVM, Decision Trees	Acc: ~75%	Regional limitation; narrow focus; 10 groups	Global applicability; comprehensive statistical validation; interpretability-enhanced ensembles
Abdalsalam et al. (2024a, b) (Feature-based)	Multi-class Classification	Attack and group classification	GTD	Text + feature fusion	Acc: ~82%	Reliance on textual features; limited interpretability	Multi-modal structured features (temporal, geographic, incident-specific); 83-class capability; operational attribution capability
Saini & Bansal (2024) (Survey)	Multi-class Attribution Overview	Review of computational techniques	Multiple	Various ML/DL approaches	Qualitative; no experiments, only trend synthesis using GTD references	No empirical evaluation; focuses on trends	Empirical validation with real-world GTD data; interpretable ensemble framework with statistical significance

Jaish-e-Mohammed and Taliban, were also assessed using SHAP values. This framework enables analysts to understand attribution reasoning, assess confidence, and identify cases warranting further investigation or manual review.

4.9.5 Computational performance analysis

SHAP computation overhead was evaluated for operational feasibility. For 100 predictions, SHAP analysis required 2.3 s and 256 MB of memory for XGBoost, 1.8 s and 192 MB for Random Forest, and 2.1 s and 220 MB for CatBoost, all rated as excellent for real-time use. The Stacking Classifier and Weighted Voting Ensemble required 8.7 and 6.4 s, respectively, with higher but still acceptable memory usage. Overall, SHAP analysis adds 2–9 s per prediction batch, making it suitable for near-real-time deployment while providing essential decision transparency.

4.9.6 Real-world implications and comparative analysis

To elucidate the operational significance and advancement of the proposed approach, Table 3 presents a comparative analysis of recent machine learning strategies in terrorism research, spanning from binary prediction tasks to complex multi-class attribution. This examination emphasizes the increase in methodological sophistication and task difficulty, and illustrates how the presented ensemble framework satisfies genuine world counter-terrorism requirements by seeking a balance of precision and recall for action-oriented intelligence and responsive effectiveness. Performance metrics are put into context based upon task difficulty, with the presented research addressing the most difficult case of distinguishing between 83 terrorist organizations while being competitively accurate and operationally pertinent. Notably, the balanced precision-recall trade-off (72.15% precision, 73.42% recall) directly supports critical counter-terrorism operations, enabling real-time threat assessment, improved resource allocation, and reliable preliminary attribution in complex operational contexts.

These findings underscore the practical value and impact of advanced machine learning in supporting global security efforts.

4.9.7 Ethical, legal, and operational risks in automated terrorism attribution

The use of automated terrorist attribution systems poses serious ethical, legal, and operational threats that need to be systematically addressed. Ethically speaking, the risk of biased algorithms is acute as a result of the restrictions posed by historical data, disparities in geographic reporting, and potential biases in inadvertent profiling on cultural or regional characteristics even more relevant considering that regional indicators account for 15.8% of attribution judgments in the current system. These biases could be perpetuating unjust outcomes or validating already entrenched inequities. From a legal standpoint, issues include the requirement of unambiguous accountability structures for automatic decisions, conformity to standards of evidence in judicial proceedings, and conformity to new AI regulatory requirements, e.g., obligatory human review and bias testing protocols. Automatic outputs need to be validly tested, since failures can lead to unfounded suspicion or intensification of conflict. Operationally, threats include over-dependence on algorithmic results, adversarial learning by terrorist organizations to dodge discovery, and susceptibility to information security threats like data poisoning and model inversion attacks. These issues are mitigated by the inclusion of explainable AI via SHAP analysis (resulting in 94% correlation with conventional measures of importance), confidence calibration for threshold-based decision processes, and human-in-the-loop functionality for every high-consequence attribution. Continuous bias monitoring, multi-source validation against traditional intelligence techniques, and the formation of ethics review boards are advised to manage deployment and address evolving challenges. These combined promote responsible, transparent, and accountable use of automated attribution systems in dynamic counter-terrorism contexts.

5 Limitations

Although this research establishes meaningful progress in terrorist group attribution, some significant limitations need to be recognized. The model is based solely on past data available in the Global Terrorism Database (GTD) for the period 1970–2020, and it is possible that such data might not reflect new terrorist strategies or the working patterns of newly established groups. This assumption assumes temporal stability of terrorist behavior a condition which might not exist as the organizations change and evolve to counter-terrorism measures. Accordingly, model accuracy could decline when faced with groups that formed subsequent to the training horizon or those showing extreme departures from past patterns, requiring recurring retraining as fresh incident data are accrued.While the GTD provides comprehensive structured records of real-world incidents, the current framework is primarily based on historical incident data and does not yet incorporate more dynamic data streams, which could further enhance responsiveness to emerging terrorist behaviors and tactics.

Further, the computational requirements of ensembles can limit deployment in resource-scoped operational environments. The architecture also makes assumptions

of single-group attribution for each incident, which could potentially oversimplify multifaceted situations such as coordinated attacks or a series of cooperating organizations.

Even with the adoption of stringent cross-validation and SHAP-based interpretability methods to reduce the risk of bias and increase transparency, the model cannot fully eliminate the possibility of inheriting historical biases embedded in the data. Additionally, the model cannot fully substitute expert human judgment in intricate cases that are defined by uncertain evidence or new operational patterns.

6 Conclusion

This study systematically advances the field of terrorism analytics by introducing and validating novel ensemble machine learning methods for multi-class terrorist group attribution. The proposed stacking classifier and weighted voting ensemble achieve 73.70% and 73.38% accuracy, respectively, in the complex task of attributing incidents to 83 terrorist groups demonstrating robust performance and operational significance in a domain where random accuracy is only 1.2%. Comprehensive comparative evaluation across twelve algorithms, supported by rigorous statistical validation and demonstrated cross-cultural applicability, confirms the superiority of ensemble approaches for modeling complex terrorist behaviors. The integration of SHAP-based interpretability addresses critical transparency requirements, enabling instance-level auditability and supporting deployment in high-stakes security contexts. The framework's balanced precision-recall performance and actionable interpretability features provide practical value for intelligence agencies, supporting automated yet explainable attribution and informing evidence-based counter-terrorism strategies. Through identifying attack type and target type as key attribution signals, the model offers insights for threat prevention and facilitates international cooperation through enhanced inter-agency intelligence sharing capabilities.

This research fills critical gaps in the literature by addressing multi-class attribution, providing systematic algorithm comparison, and integrating explainability into operational models.

6.1 Future research directions

Future research will prioritize several directions to further strengthen attribution accuracy and resilience. These include developing real-time adaptation mechanisms, advancing federated learning for secure multi-institutional collaboration, and enhancing adversarial robustness. Integration of multi-modal data, particularly more dynamic data streams beyond structured historical incident records-remains a key objective to improve responsiveness to emerging terrorist behaviors and tactics.

In sum, this study delivers a significant methodological advance in multi-class terrorist group attribution by introducing and statistically validating novel ensemble approaches that provide transparent, operationally viable, and scalable solutions-setting a new benchmark for the field and offering a strong foundation for future innovation in security informatics.

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Author contributions All authors contributed to the study, and design. Conception, material preparation, data collection and analysis were performed by GM. The first draft of the manuscript was written by GM, Methodology- GM, LC, YP & JW; Validation -all authors; Writing original draft preparation, all authors; Writing and editing, GM; Review; all authors and finally all authors commented on previous versions, read and approved the final manuscript.

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Data availability The original terrorism incident data utilized in this study were obtained from the Global Terrorism Database (GTD), which is publicly available at https://www.start.umd.edu/gtd/ (accessed January 2025) and also accessible via Zenodo at https://zenodo.org/record/4719251. The GTD is maintained by the National Consortium for the Study of Terrorism and Responses to Terrorism (START), University of Maryland. In accordance with GTD usage terms, the raw dataset cannot be redistributed directly by the authors. However, the specific preprocessing pipeline and feature engineering code used in this analysis are available from the corresponding author upon reasonable request. The complete preprocessing methodology is fully documented in Section 3 of the manuscript to ensure transparency and reproducibility.

Declarations

Competing interest The authors declare no competing interests.

Human and animal participations Not applicable.

Informed consent All authors have read and agreed to the published version of the manuscript.

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References

Abdalsalam M, Li C, Dahou A, Kryvinska N (2024b) Terrorism group prediction using feature combination and BiGRU with self-attention mechanism. PeerJ Comput Sci 10:e2252

Abdalsalam M, Li C, Dahou A, Kryvinska N (2024) Terrorism attack classification using machine learning: the effectiveness of using textual features extracted from GTD dataset. CMES-Computer Modeling in Engineering & Sciences, 138(2).

Abdelhamid AA, Towfek SK, Khodadadi N, Alhussan AA, Khafaga DS, Eid MM, Ibrahim A (2023) Waterwheel plant algorithm: a novel metaheuristic optimization method. Processes 11(5):1502

Adadi A, Berrada M (2018) Peeking inside the black-box: a survey on explainable artificial intelligence (XAI). IEEE Access 6:52138–52160

Adigwe CS, Mayeke NR, Olabanji SO, Okunleye OJ, Joeaneke P, Olaniyi OO (2024) The evolution of terrorism in the digital age: Investigating the adaptation of terrorist groups to cyber technologies for recruitment, propaganda, and cyberattacks. Propag Cyberattack 24(3):289–306

Agarwal S, Sureka A (2015) Using knn and svm based one-class classifier for detecting online radicalization on twitter. In: Distributed Computing and Internet Technology: 11th International Conference, ICDCIT 2015, Bhubaneswar, India, February 5–8, 2015. Proceedings 11 (pp. 431–442).

Ahmad S, Asghar MZ, Alotaibi FM, Awan I (2019) Detection and classification of social media-based extremist affiliations using sentiment analysis techniques. HCIS 9:1–23

Alazab M, Soman KP, Srinivasan S, Venkatraman S, Pham VQ (2023) Deep learning for cyber security applications: a comprehensive survey. Authorea Preprints.

ALfatih M, Li C, Saadalla NE (2019) Prediction of groups responsible for terrorism attack using tree based models. In: Proceedings of the 2019 International Conference on Artificial Intelligence and Computer Science, pp 320–324

Alzakari SA, Alhussan AA, Qenawy AST et al (2025) Early detection of potato disease using an enhanced convolutional neural network-long short-term memory deep learning model. Potato Res 68:695–713

Ariyo O, Agbaje MO, Oyebola A, Izang AA (2024) Comparative examination of machine learning models for terrorist activity prediction. The Journal Of Computational Science And Engineering, vol 2(6).

Atoum MS, Alarood AA, Alsolmi EA, Obeidat A, Alazab M (2025) Predictive analysis of global terrorist attacks using lexical patterns across multiple datasets. Expert Syst 42(1):e13808

Black, R., Busby, J., Dabelko, G. D., De Coning, C., Maalim, H., McAllister, C.,. & Staudenmann, J. A. (2022).

Bolshibayeva AK, Rakhmetulayeva S, Kulbayeva AK (2023) Machine learning methods to detect terrorist financing. In: Proceedings of the 8th International Conference on Digital Technologies in Education, Science and Industry (DTESI 2023), Almaty, Kazakhstan, (Vol. 3680, pp. 1–12).

Bridgelall R (2022) An application of natural language processing to classify what terrorists say they want. Soc Sci 11(1):23

Buffa C, Sagan V, Brunner G, Phillips Z (2022) Predicting terrorism in Europe with remote sensing, spatial statistics, and machine learning. ISPRS Int J Geo Inf 11(4):211

Derrac J, García S, Molina D, Herrera F (2011) A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. Swarm Evol Comput 1(1):3–18

Devlin J, Chang MW, Lee K, Toutanova K (2019) Bert: pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, vol 1 pp 4171–4186

Ding F, Ge Q, Jiang D, Fu J, Hao M (2017) Understanding the dynamics of terrorism events with multiple-discipline datasets and machine learning approach. PLoS ONE 12(6):e0179057

El-Kenawy ESM, Khodadadi N, Mirjalili S, Abdelhamid AA, Eid MM, Ibrahim A (2024) Greylag goose optimization: nature-inspired optimization algorithm. Expert Syst Appl 238:122147

El-Kenawy ESM, Alhussan AA, Khodadadi N et al (2025) Predicting potato crop yield with machine learning and deep learning for sustainable agriculture. Potato Res 68:759–792

El-kenawy EM (2025) A review of machine learning models for predicting air quality in urban areas. Metaheurist Optimiz Rev 3(2):33–46

El-kenawy ESM, Abutarboush HF, Mohamed AW, Ibrahim A (2021) Advance artificial intelligence technique for designing double T-shaped monopole antenna. Computers, Materials & Continua, 69(3).

Elshabrawy M (2025) A review on waste management techniques for sustainable energy production. Metaheurist Optimiz Rev 3(2):47–58

Feng Y, Wang D, Yin Y, Li Z, Hu Z (2020) An XGBoost-based casualty prediction method for terrorist attacks. Complex Intell Syst 6:721–740

Fraiwan M (2022) Identification of markers and artificial intelligence-based classification of radical Twitter data. Appl Comput Inform 18(2):1–13

Gundabathula VT, Vaidhehi V (2018) An efficient modelling of terrorist groups in India using machine learning algorithms. Indian J Sci Technol 11(15):1–10

Haghani M, Kuligowski E, Rajabifard A, Lentini P (2022) Fifty years of scholarly research on terrorism: intellectual progression, structural composition, trends and knowledge gaps of the field. Int J Dis Risk Reduct 68:102714

Jaspersen JG, Montibeller G (2020) On the learning patterns and adaptive behavior of terrorist organizations. Eur J Oper Res 282(1):221–234

Khan FA, Li G, Khan AN, Khan QW, Hadjouni M, Elmannai H (2023) AI-driven counter-terrorism: enhancing global security through advanced predictive analytics. IEEE Access 11:135864–135879

Lu SY, Zhu Z, Tang Y, Zhang X, Liu X (2025a) CTBViT: A novel ViT for tuberculosis classification with efficient block and randomized classifier. Biomed Signal Process Control 100:106981

Lu SY, Zhang YD, Yao YD (2025b) A regularized transformer with adaptive token fusion for Alzheimer's disease diagnosis in brain magnetic resonance images. Eng Appl Artif Intell 155:111058

Lundberg SM, Erion G, Chen H, DeGrave A, Prutkin JM, Nair B, Lee SI (2020) From local explanations to global understanding with explainable AI for trees. Nat Mach Intell 2(1):56–67

Lundberg SM, Lee S-I (2017) A unified approach to interpreting model predictions. In: Proceedings of the 31st International Conference on Neural Information Processing Systems, pp 4765–4774

Mahmood N, Ghani Khan MU (2022) Prediction of extremist behaviour and suicide bombing from terrorism contents using supervised learning. Computers, Materials & Continua, 70(3).

Maniszewska K (2024) Countering terrorism: key challenges and proposed solutions. In: Towards a New Definition of Terrorism: Challenges and Perspectives in a Shifting Paradigm (pp. 65–82)

Mohamed A, Mahmood S (2025) K-nearest neighbors approach to analyze and predict air quality in Delhi. J Artif Intell Metaheurist. https://doi.org/10.54216/JAIM.090104

Montasari R (2024) Addressing ethical, legal, technical, and operational challenges in counter-terrorism with machine learning: recommendations and strategies. In: Cyberspace, Cyberterrorism and the International Security in the Fourth Industrial Revolution: Threats, Assessment and Responses (pp. 199- 226).

Oketch PO, Mukisa MT, Ratemo MC (2019) An evaluation of hybrid machine learning classifier models for identification of terrorist groups in the aftermath of an attack. Int Res J Eng Technol (IRJET) 6(9):1856–1862

Rahman MA, Islam MR, Islam MT, Sadi MS (2022) Explainable artificial intelligence for cybersecurity incident classification. Comput Secur 116:102667

Renard T (2021) The Evolution of Counter-Terrorism Since 9/11: Understanding the Paradigm Shift in Liberal Democracies (1st ed.). Routledge. https://doi.org/10.4324/9781003188018

Rigutto C, Sapara AO, Agyapong VI (2021) Anxiety, depression and posttraumatic stress disorder after terrorist attacks: a general review of the literature. Behav Sci 11(10):140

Saif H, Dickinson T, Kastler L, Fernandez M, Alani H (2017) A semantic graph-based approach for radicalization detection on social media. In: The Semantic Web: 14th International Conference, ESWC 2017, Portorož, Slovenia, May 28–June 1, 2017, Proceedings, Part I 14 (pp. 571–587).

Saini JK, Bansal D (2024) Computational techniques to counter terrorism: a systematic survey. Multimedia Tools Appl 83(1):1189–1214

Sarker A, Chakraborty P, Sha SS, Khatun M, Hasan MR, Banerjee K (2020) Improvised technique for analyzing data and detecting terrorist attack using machine learning approach based on twitter data. J Comput Commun 8(7):50–62

Gaber ShK, Singla MK (2025) Predictive Analysis of Groundwater Resources Using Random Forest Regression. Journal of Artificial Intelligence and Metaheuristics. pp 11–19.

Sharif W, Mumtaz S, Shafiq Z, Riaz O, Ali T, Husnain M, Choi GS (2019) An empirical approach for extreme behavior identification through tweets using machine learning. Appl Sci 9(18):3723

Siddiqui JA, Yuhaniz SS, Shaikh GM, Soomro SA, Mahar ZA (2024) Fine-grained multilingual hate speech detection using explainable AI and transformers. IEEE Access 12:143177–143192

Singh S, Kumar A, Sharma S (2023) Interpretability in AI-driven threat assessment: a SHAP-based approach. J Sec Inform 12(2):45–59

Singh K, Chaudhary AS, Kaur P (2019). A machine learning approach for enhancing defence against global terrorism. In: 2019 Twelfth International Conference on Contemporary Computing (IC3) (pp. 1–5). IEEE.

Torregrosa J, Thorburn J, Lara-Cabrera R, Camacho D, Trujillo HM (2020) Linguistic analysis of pro-ISIS users on Twitter. Behav Sci Terror Politic Aggr 12(3):171–185

Tutun S, Khasawneh MT, Zhuang J (2017) New framework that uses patterns and relations to understand terrorist behaviors. Expert Syst Appl 78:358–375

Uddin MI, Zada N, Aziz F, Saeed Y, Zeb A, Ali Shah SA, Al- Khasawneh MA, Mahmoud M (2020) Prediction of future terrorist activities using deep neural networks. Complexity 2020(1):1373087

Yang Z, Dai Z, Yang Y, Carbonell J, Salakhutdinov RR, Le QV (2019) XLNet: generalized autoregressive pretraining for language understanding, vol 32

Zhu Z, Ren Z, Lu S, Wang S, Zhang Y (2023) DLBCNet: a deep learning network for classifying blood cells. Big Data Cogn Comput 7(2):75

Zhu Z, Liu L, Free RC, Anjum A, Panneerselvam J (2024) OPT-CO: optimizing pre-trained transformer models for efficient COVID-19 classification with stochastic configuration networks. Inf Sci 680:121141

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